

QMEA (Inference about a Population)

1. When the population standard deviation σ is unknown, which distribution is generally used for inference about a population mean?

- A. Standard normal distribution
- B. t distribution
- C. Chi-square distribution
- D. Binomial distribution

2. A researcher calculates a 95% confidence interval for μ as: (48.2, 53.6)

Which interpretation is most appropriate?

- A. 95% of observations lie between 48.2 and 53.6
- B. The sample mean has a 95% chance of being in this interval
- C. The interval provides plausible values for the population mean
- D. The population mean changes from sample to sample

3. Which distribution is used for inference about the variance of a normally distributed population?

- A. t distribution
- B. Standard normal distribution
- C. Chi-square distribution
- D. F distribution only

4. Which assumption is particularly important for inference about a population variance using the chi-square distribution?

- A. The population is approximately normal
- B. The sample size must always exceed 100
- C. The population mean must be known
- D. The population variance must already be known

5. Which situation is most appropriate for inference about a population proportion?

- A. Estimating the average income of households
- B. Estimating the variability of product weights
- C. Estimating the percentage of voters who support a policy
- D. Estimating the average waiting time at a hospital

6. A hospital administrator wants to determine whether the average waiting time of patients is less than 30 minutes. A random sample of 40 patients is selected. The population standard deviation is unknown.

Which statistical test is most appropriate?

- A. One-sample z-test for a mean
- B. One-sample t-test for a mean
- C. Chi-square test for a variance
- D. One-proportion z-test

7. A factory produces metal rods. The manager is concerned that the variability in rod length has increased above the acceptable variance of 0.04 cm^2 . A random sample of 30 rods is selected.

Which test is most appropriate?

- A. One-sample t-test
- B. One-proportion z-test
- C. Chi-square test for a population variance
- D. z-test for a population mean

8. A filling machine is designed so that the standard deviation of the amount filled into bottles is no more than 2 mL. The quality-control manager wants to determine whether the machine has become less consistent.

Which test should be used?

- A. One-sample t-test for the mean
- B. Chi-square test for the variance
- C. One-proportion z-test
- D. Paired t-test

9. A company is satisfied with the average weight of its packages but is concerned that package weights have become more dispersed than before. It wants to test whether the population variance exceeds 16 grams^2 .

Which procedure should be used?

- A. One-sample t-test
- B. One-sample z-test
- C. Chi-square test for variance
- D. One-proportion z-test

10. A bank claims that at least 90% of its customers are satisfied with its mobile banking service. A random sample of 300 customers is surveyed, and each customer answers either "satisfied" or "not satisfied."

Which test is most appropriate?

- A. One-sample t-test
- B. Chi-square test for variance
- C. One-proportion z-test
- D. Two-sample t-test

11. Historically, the standard deviation of delivery times for a company has been 4 days. A manager wants to determine whether the average delivery time has decreased from 20 days. A random sample of 64 deliveries has a mean delivery time of 18.8 days.

Which test should be used?

- A. One-sample z-test for a population mean
- B. One-sample t-test for a population mean
- C. Chi-square test for a population variance
- D. One-proportion z-test

12. A manufacturer wants to determine whether the variability in package weights has increased. The acceptable population variance is 16 grams². A random sample of 30 packages is selected from a normally distributed population.

Which test should be used?

- A. One-sample z-test for a population mean
- B. One-sample t-test for a population mean
- C. Chi-square test for a population variance
- D. One-proportion z-test

13. A machine produces metal rods. The population standard deviation of rod length is known to be 0.5 cm. A random sample of 36 rods is selected. The manager wants to determine whether the mean rod length differs from 50 cm.

Which test should be used?

- A. Chi-square test because the standard deviation is given
- B. One-sample t-test because a sample is being used
- C. One-sample z-test for a population mean
- D. One-proportion z-test

14. A factory historically produces bottles with an average fill of 500 mL and a standard deviation of 4 mL. After maintenance, a sample of 50 bottles has a mean of 501.3 mL and a sample standard deviation of 5.1 mL. The manager wants to determine whether the average fill has changed.

Which test should be used?

- A. One-sample z-test for a population mean
- B. One-sample t-test for a population mean
- C. Chi-square test for a population variance
- D. One-proportion z-test

15. A university surveys 300 students. The average number of hours studied per week is 14.2, with a sample standard deviation of 6.1 hours. Of the 300 students, 174 study at least 15 hours per week.

The university wants to determine whether more than half of all students study at least 15 hours per week.

Which test should be used?

- A. One-sample t-test for a population mean
- B. One-sample z-test for a population mean
- C. Chi-square test for a population variance
- D. One-proportion z-test

16. A quality-control engineer knows that the population standard deviation of package weight was historically 2 grams. After changing suppliers, a sample of 40 packages has a sample standard deviation of 3 grams.

Which research question would require a chi-square test?

- A. Has the average package weight changed?
- B. Has the proportion of underweight packages increased?
- C. Has the variability in package weight increased?
- D. Is the mean package weight above the target?

17. A hospital collects waiting-time data from 100 patients. The sample mean is 26 minutes, the sample standard deviation is 12 minutes, and 78 patients waited less than 30 minutes.

Which pairing is correct?

- A. "Is the mean waiting time below 30 minutes?" → proportion z-test
- B. "Do more than 75% wait less than 30 minutes?" → t-test
- C. "Is the variability in waiting times greater than a specified value?" → chi-square test
- D. "Is the mean waiting time below 30 minutes?" → chi-square test

18. A university introduced a new academic support program. Before the program was introduced, students studied an average of 12 hours per week, with a population standard deviation of 4 hours.

After the program was introduced, a random sample of 200 students was surveyed. The sample had an average study time of 13.1 hours per week with a sample standard deviation of 4.5 hours. Among these students, 124 studied more than 12 hours per week.

The university wants to determine whether a majority of students now study more than the previous average of 12 hours per week.

Which statistical test and hypotheses are most appropriate?

A. One-sample z-test for a population mean

$$H_0: \mu = 12, H_1: \mu > 12$$

B. One-sample t-test for a population mean

$$H_0: \mu = 12, H_1: \mu > 12$$

C. One-proportion z-test

$$H_0: p = 0.50, H_1: p > 0.50$$

D. Chi-square test for a population variance

$$H_0: \sigma^2 = 16, H_1: \sigma^2 > 16$$

19. A university surveys 300 students. The average amount of sleep is 6.8 hours per night, with a sample standard deviation of 1.2 hours. Of the 300 students, 132 sleep at least 7 hours per night.

The university wants to determine whether students sleep an average of less than 7 hours per night.

Which test should be used?

A. One-sample t-test for a population mean

B. One-sample z-test for a population mean

C. Chi-square test for a population variance

D. One-proportion z-test

20. A university surveys 400 students. Students spend an average of 3.8 hours per day using their smartphones, with a sample standard deviation of 1.5 hours. Of the 400 students, 270 use their smartphones for more than 3 hours per day. The university wants to investigate whether students spend an average of more than 3.5 hours per day using their smartphones. Which test should be used?

A. One-proportion z-test because 270 out of 400 students exceed 3 hours

B. One-sample z-test because the sample size is large

C. One-sample t-test because the question concerns a mean and σ is unknown

D. Chi-square test because the sample standard deviation is given

21. A hospital studies blood pressure in 250 patients. The sample mean systolic blood pressure is 126 mmHg, with a sample standard deviation of 18 mmHg. Of the 250 patients, 160 have systolic blood pressure below 130 mmHg.

A researcher wants to determine whether the population mean systolic blood pressure differs from 130 mmHg.

Which combination is correct?

- A. One-proportion z-test; $H_1: p \neq 0.50$
- B. One-sample t-test; $H_1: \mu \neq 130$
- C. One-sample z-test; $H_1: \mu < 130$
- D. Chi-square test; $H_1: \sigma^2 \neq 130$

22. A quality-control manager wants to investigate three questions:

1. Is the average bottle volume different from 500 mL?
2. Has the variability of bottle volume increased?
3. Is the percentage of defective bottles greater than 2%?

Which parameters should be investigated, respectively?

- A. p, μ, σ^2
- B. μ, σ^2, p
- C. σ^2, p, μ
- D. μ, p, σ^2